

# Ally Stauffer

+1 (925) 357-7542 | astauf02@calpoly.edu | linkedin.com/in/allystauffer | github.com/allysonstauffer

## EDUCATION

### Bachelor of Science, Computer Science | Minor in Mathematics | 3.61 GPA

*Expected May 2028*

California Polytechnic State University, San Luis Obispo

*San Luis Obispo, CA*

Relevant Coursework: Data Structures (Java), Object-Oriented Programming & Design (Java), Algorithms, Database Management Systems, Systems Programming, Artificial Intelligence, Data Science

### UC Davis COSMOS — Quantum Mechanics & Applications to Nanotechnology (Certificate)

*Summer 2023*

## WORK EXPERIENCE

### Software Engineer

*Aug 2026 – Present*

Cal Poly Noyce School of Applied Computing

*San Luis Obispo, CA*

- Selected as the primary student Software Engineer to build an infrastructure-monitoring and data-visualization platform for Cal Poly's \$3M NVIDIA AI Factory, covering NVIDIA DGX B200 GPU systems, compute, networking, storage, and power infrastructure.
- Designing telemetry pipelines and stakeholder-facing dashboards for workload, utilization, capacity, and energy metrics, using Claude Code and Cursor to accelerate prototyping while critically reviewing LLM-generated code before merge.

### Software Engineering Intern

*Jun 2026 – Aug 2026*

Kestrix

*London, England, UK*

- Built a Python to BigQuery to Looker Studio scoring pipeline that turned scattered flight-validation signals into an hourly, property-level ops dashboard, so incomplete coverage is caught in hours, not days.
- Wrote 47 pytest cases over 11 fixtures with expected outputs, covering property scoring, remediation (no action / re-upload / reflight), and env loading, then mirrored that logic into BigQuery SQL.
- Investigated validation signals across 7 PostgreSQL/BigQuery entities and built SQLAlchemy/Folium maps of planned vs collected imagery so ops could see which properties still needed recapture.
- Used Claude Code and GitHub Copilot to draft SQL and Python, reviewing outputs for edge cases before shipping a Looker work-queue dashboard ops could filter and export.

### Software Developer, Mustang Maps

*Sep 2025 – Jun 2026*

CodeBox

*San Luis Obispo, CA*

- Production React Native campus navigator for Cal Poly (161+ downloads, 8-person team) using team-built A\* over a GeoJSON walking graph (~5.7k nodes, 6.3k edges, ~98 km of paths, 400+ buildings).
- Shipped a MapLibre class-zones layer over campus GeoJSON — 92 floor-plan zones across 45 buildings, covering 4,271 classrooms — so students can tap a building and see which wing and floor a room sits on.
- Wired GO/Exit navigation-mode state for turn-by-turn on top of that A\* graph; shipped in a production React Native app.

### Teaching Assistant

*May 2025 – Aug 2025*

Girls Who Code

*Remote / Bay Area*

- Taught 100+ high school students game design with p5.js (JavaScript), covering mechanics, design, and creative coding; led daily advisory focused on engagement and inclusivity.
- Ran 1:1 debugging and project feedback in office hours; managed attendance, surveys, and accommodations across multiple time zones.

## PROJECTS

### LocLens | React Native, Expo, React, TypeScript, Node.js, Express, PostgreSQL

*May 2026 – Present*

- Building a full-stack field-documentation platform for AEC teams: Expo mobile capture with offline queueing/retry, multi-attribute photo metadata, and structured project organization.
- Developed a React/Express/PostgreSQL backend with RBAC, metadata audit history, bulk tagging, and ZIP/CSV project exports for unreliable field connectivity workflows.

### Sightline | Next.js, TypeScript, React Three Fiber, Tailwind, Gemini API

*Sep 2026*

- Built a camera-first room editor that turns 6–12 photos into an editable 2D/3D plan; Gemini handles perception, while a deterministic engine encodes 12 spatial principles and 5 Neufert clearances to create 3 fit-verified layouts with a 4-state per-piece verdict.
- Built a parallel recommendation pipeline across 8 retailers (up to 34 listings/category), ranking on price, vibe, and parsed WxDxH; unverified dimensions never get placed, and pieces over 45% of room area or within 1 ft of a wall are rejected.

## TECHNICAL SKILLS

**Languages** | Java, JavaScript, TypeScript, Python, SQL, C/C++, HTML/CSS

**Frameworks** | React, React Native, Next.js, Node.js/Express

**Data & Tools** | PostgreSQL, BigQuery, Git, Pandas, Looker Studio

**AI-Assisted Dev** | Claude Code, GitHub Copilot, Warp, Cursor, ChatGPT/Codex